

UNIT1

4 / 6 Marks:

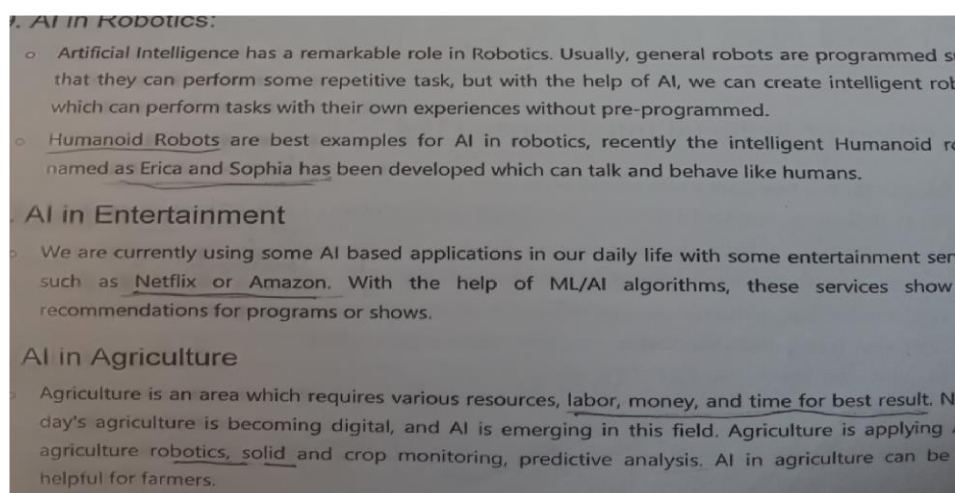
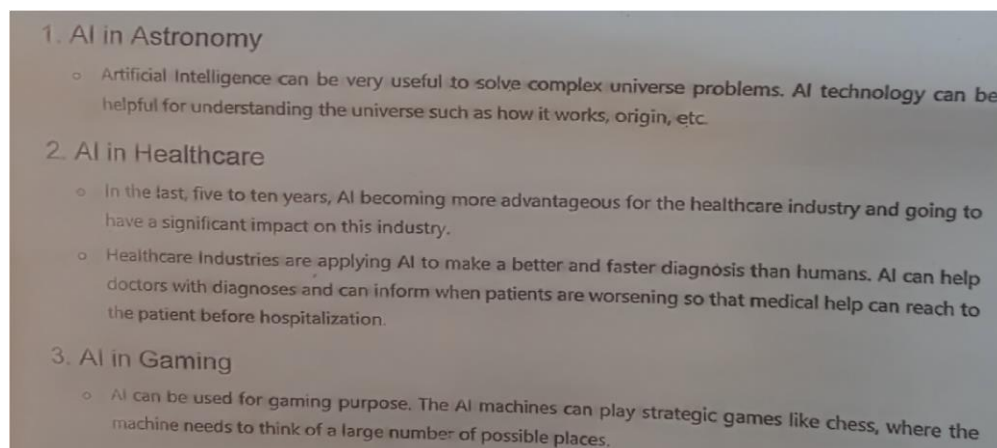
1. Write a note on “Acting humanly, the Turing test approach”. ?

Acting humanly: The Turing Test approach o Test proposed by Alan Turing in 1950 o The computer is asked questions by a human interrogator.

The computer passes the test if a human interrogator, after posing some written questions, cannot tell whether the written responses come from a person or not. Programming a computer to pass, the computer need to possess the following capabilities:

- Natural language processing to enable it to communicate successfully in English.
- Knowledge representation to store what it knows or hears
- Automated reasoning to use the stored information to answer questions and to draw new conclusions
- Machine learning to adapt to new circumstances and to detect and extrapolate
- patterns to pass the complete Turing Test, the computer will need
- Computer vision to perceive the objects
- Robotics to manipulate objects and move about

6. Explain any six AI applications.?



7. Explain any six state of the art AI applications.?

Autonomous control: The ALVINN computer vision system was trained to steer a car to keep it following a lane. It was placed in CMU's NAVLAB computer-controlled minivan and used to navigate across the United States.

Game playing: IBM's Deep Blue became the first computer program to defeat the world champion in a chess match when it bested Garry Kasparov by a score of 3.5 to 2.5 in an exhibition match (Goodman and Keene, 1997).

Diagnosis: Medical diagnosis programs based on probabilistic analysis have been able to perform at the level of an expert physician in several areas of medicine.

Autonomous planning and scheduling: A hundred million miles from Earth, NASA's Remote Agent program became the first on-board autonomous planning program to control the scheduling of operations for a spacecraft (Jonsson et al., 2000). Remote Agent generated plans from high-level goals specified from the ground, and it monitored the operation of the spacecraft as the plans were executed—detecting, diagnosing, and recovering from problems as they occurred.

Robotics: Many surgeons now use robot assistants in microsurgery. HipNav (DiGioia et al., 1996) is a system that uses computer vision techniques to create a three-dimensional model of a patient's internal anatomy and then uses robotic control to guide the insertion of a hip replacement prosthesis.

Language understanding and problem solving: PROVERB (Littman et al., 1999) is a computer program that solves crossword puzzles better than most humans, using constraints on possible word fillers, a large database of past puzzles, and a variety of information sources including dictionaries and online databases such as a list of movies and the actors that appear in them.

11. How do you specify a task environment using PEAS description? Explain with an example.

We must think about task environments, which are essentially the "problems" to which rational agents are the "solutions." Specifying the task environment The rationality of the simple vacuum-cleaner agent, needs specification of

- the performance measure
- the environment
- the agent's actuators and sensors.

PEAS : All these are grouped together under the heading of the task environment. We call this the PEAS (Performance, Environment, Actuators, Sensors) description. In designing an agent, the first step must always be to specify the task environment as fully as possible.

Agent Type	Performance Measure	Environments	Actuators	Sensors
Taxi driver	Safe: fast, legal, comfortable trip, maximize profits	Roads, other traffic, pedestrians, customers	Steering, accelerator, brake, Signal, horn, display	Cameras, sonar, Speedometer, GPS, Odometer, engine sensors, keyboards, accelerometer

Figure 1.5 PEAS description of the task environment for an automated taxi.

15. Explain simple reflex agent with a neat diagram.

The simplest kind of agent is the simple reflex agent. These agents select actions on the basis of the current percept, ignoring the rest of the percept history. For example, the vacuum agent whose agent function is a simple reflex agent, because its decision is based only on the current location and on whether that contains dirt.

- Select action on the basis of only the current percept. E.g. the vacuum-agent
- Large reduction in possible percept/action situations(next page).
- Implemented through condition-action rules If dirty then suck

A Simple Reflex Agent: Schema

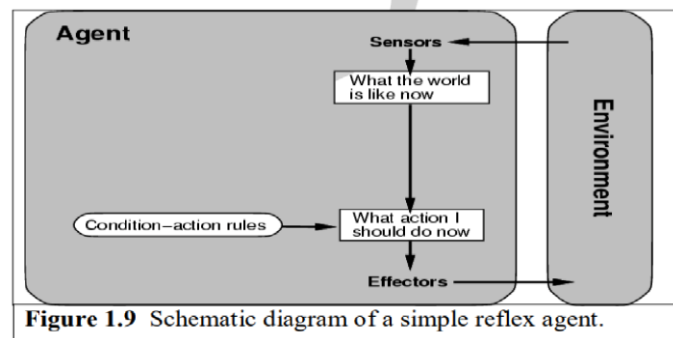


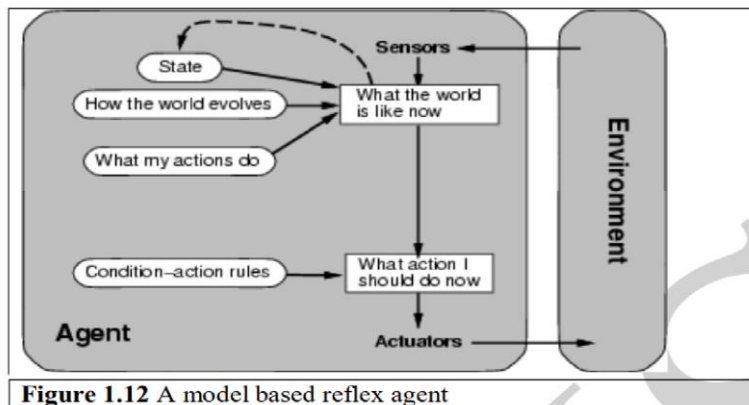
Figure 1.9 Schematic diagram of a simple reflex agent.

Characteristics

- Only works if the environment is fully observable.
- Lacking history, easily get stuck in infinite loops
- One solution is to randomize actions

16. Briefly explain model based reflex agent with a diagram.?

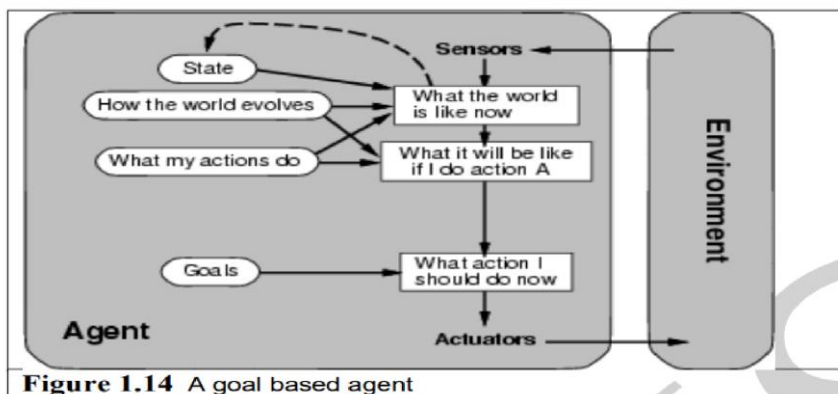
The most effective way to handle partial observability is for the agent to keep track of the part of the world it can't see now. That is, the agent should maintain some sort of internal state that depends on the percept history and thereby reflects at least some of the unobserved aspects of the current state. Updating this internal state information as time goes by requires two kinds of knowledge to be encoded in the agent program. First, we need some information about how the world evolves independently of the agent-for example, that an overtaking car generally will be closer behind than it was a moment ago. Second, we need some information about how the agent's own actions affect the world-for example, that when the agent turns the steering wheel clockwise, the car turns to the right or that after driving for five minutes northbound on the freeway one is usually about five miles north of where one was five minutes ago. This knowledge about "how the world working - whether implemented in simple Boolean circuits or in complete scientific theories-is called a model of the world. An agent that uses such a MODEL-BASED model is called a model-based agent.



17. Briefly explain goal based reflex agent with a diagram?

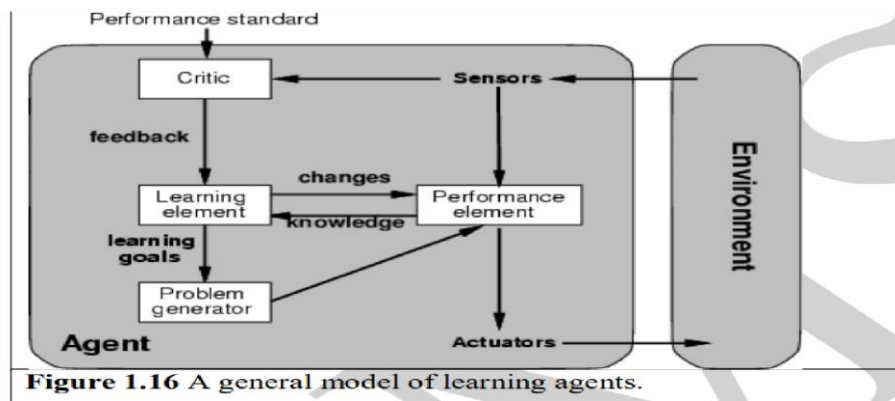
Knowing about the current state of the environment is not always enough to decide what to do. For example, at a road junction, the taxi can turn left, turn right, or go straight on. The correct decision depends on where the taxi is trying to get to. In other words, as well as a current state description, the agent needs some sort of goal information that describes situations that are desirable—for example, being at the passenger's destination. The agent program can combine this with information about the results of possible actions (the same information as was used to update internal state in the reflex agent) in order to choose actions that achieve the goal.

Figure shows the goal-based agent's structure



19. Explain learning agent with a diagram?

All agents can improve their performance through learning. A learning agent can be divided into four conceptual components, as shown in Figure 1.15. The most important distinction is between the learning element, which is responsible for making improvements, and the performance element, which is responsible for selecting external actions. The performance element is what we have previously considered to be the entire agent: it takes in percepts and decides on actions. The learning element uses feedback from the critic on how the agent is doing and determines how the performance element should be modified to do better in the future. The last component of the learning agent is the problem generator. It is responsible for suggesting actions that will lead to new and informative experiences. But if the agent is willing to explore a little, it might discover much better actions for the long run. The problem generator's job is to suggest these exploratory actions. This is what scientists do when they carry out experiments.



UNIT II

4/6 Marks:

5. Explain BFS algorithm with example ?

Or

16. Explain Breadth First Search. Also discuss its performance.?

Breadth-first Search:

- Breadth-first search is the most common search strategy for traversing a tree or graph. This algorithm searches breadthwise in a tree or graph, so it is called breadth-first search.
- BFS algorithm starts searching from the root node of the tree and expands all successor node at the current level before moving to nodes of next level.
- The breadth-first search algorithm is an example of a general-graph search algorithm.
- Breadth-first search implemented using FIFO queue data structure.

Advantages:

- BFS will provide a solution if any solution exists.
- If there are more than one solutions for a given problem, then BFS will provide the minimal solution which requires the least number of steps.

Disadvantages:

- It requires lots of memory since each level of the tree must be saved into memory to expand the next level.
- BFS needs lots of time if the solution is far away from the root node.

10. Explain Bidirectional search algorithm with example.?

Bidirectional search algorithm runs two simultaneous searches, one from initial state called as forward-search and other from goal node called as backward-search, to find the goal node. Bidirectional search replaces one single search graph with two small subgraphs in which one starts the search from an initial vertex and other starts from goal vertex. The search stops when these two graphs intersect each other.

Bidirectional search can use search techniques such as BFS, DFS, DLS, etc.

Advantages:

- Bidirectional search is fast.

- Bidirectional search requires less memory

Disadvantages:

- Implementation of the bidirectional search tree is difficult.
- In bidirectional search, one should know the goal state in advance.

Example:

In the below search tree, bidirectional search algorithm is applied. This algorithm divides one graph/tree into two sub-graphs. It starts traversing from node 1 in the forward direction and starts from goal node 16 in the backward direction.

The algorithm terminates at node 9 where two searches meet.

Completeness: Bidirectional Search is complete if we use BFS in both searches.

Time Complexity: Time complexity of bidirectional search using BFS is $O(bd)$.

Space Complexity: Space complexity of bidirectional search is $O(bd)$.

Optimal: Bidirectional search is Optimal.

12.Explain the vacuum world problem.?

The vacuum world problem is a classic and fundamental example in the field of artificial intelligence (AI) that is used to illustrate the concept of rational agents and their interaction with the environment. The problem is designed to demonstrate how an autonomous agent can perceive and act upon its environment to achieve a specific goal.

In the vacuum world problem, an autonomous vacuum cleaner agent is placed in a grid-like environment. The environment consists of discrete squares, some of which are clean and others are dirty. The agent is capable of perceiving the state of the squares and can perform actions to clean the dirty squares. The objective of the agent is to clean all the dirty squares in the environment.

The vacuum world problem highlights several key concepts in AI, including:

- 1. Agent-Environment Interaction:** The problem demonstrates the interaction between the agent and its environment. The agent's actions, such as moving and cleaning, affect the state of the environment, and the environment's state influences the agent's perception and decision-making.
- 2. Perception and Action:** The agent perceives the state of the squares in the environment as clean or dirty. Based on its perception, the agent decides on actions to perform, such as moving to a new square or cleaning a dirty square.
- 3. Rational Behavior:** The problem emphasizes the concept of rational behavior, where the agent aims to achieve its goal of cleaning all the dirty squares based on its perception of the environment and its actions.
- 4. Goal-Directed Behavior:** The agent's goal is explicitly defined as cleaning all the dirty squares, illustrating goal-directed behavior in an AI context.

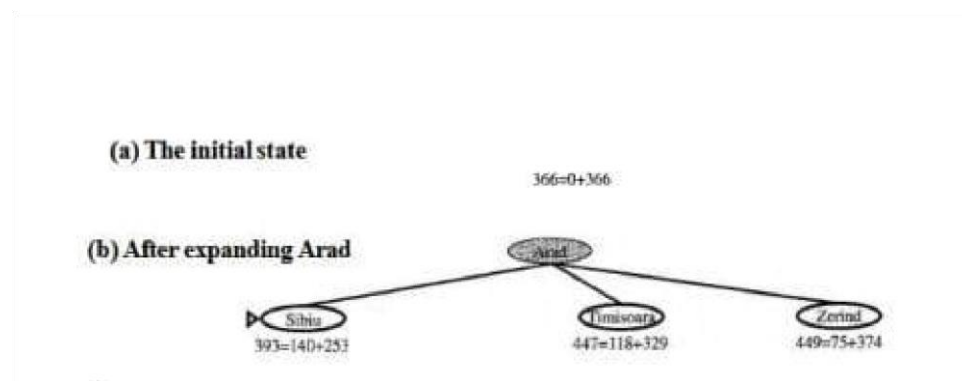
The vacuum world problem serves as a foundational example in AI to demonstrate the principles of agent-environment interaction, perception, action, and rational decision-making. It provides a simple yet effective scenario for understanding how autonomous agents can operate in dynamic environments to achieve specific objectives.

20. Explain A* algorithm. Consider the map of Romania and the hSLD value table given. Use A* algorithm to find a path from Arad to Bucharest.

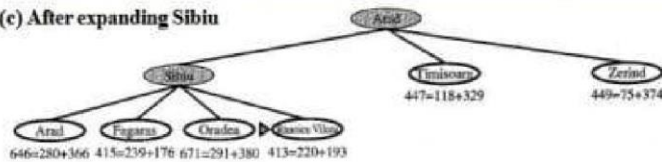
A* search is a popular algorithm used to find the shortest path between two points. It works by evaluating nodes based on two costs: $g(n)$, the cost to reach the node, and $h(n)$, the estimated cost to get from the node to the goal. The algorithm chooses the node with the lowest combined cost, $g(n) + h(n)$, to explore next. This strategy is not only reasonable but also guarantees the shortest path if the heuristic function $h(n)$ meets certain conditions.

The optimality of A* search relies on the use of an admissible heuristic, which never overestimates the cost to reach the goal. Admissible heuristics are optimistic, thinking the cost is less than it actually is. Since $g(n)$ is the exact cost to reach a node, the combined cost $f(n) = g(n) + h(n)$ never overestimates the true cost of a solution. A simple example of an admissible heuristic is the straight-line distance between two points, which is always an underestimate.

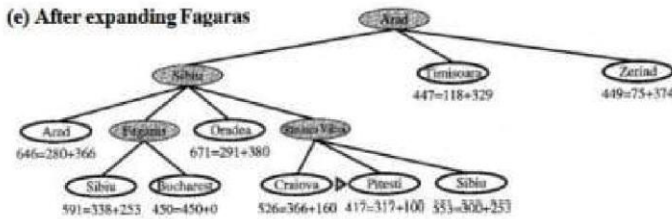
A* search is used with the TREE-SEARCH algorithm to find the shortest path. In this example, the algorithm is used to find the shortest path to Bucharest. The values of $g(n)$ are computed from the step costs, and the values of $h(n)$ are given as the straight-line distance. The algorithm chooses the node with the lowest combined cost to explore next. In this case, Bucharest appears on the fringe but is not selected because its cost is higher than that of Pitesti. The algorithm will not settle for a solution that costs more than the potential solution through Pitesti.



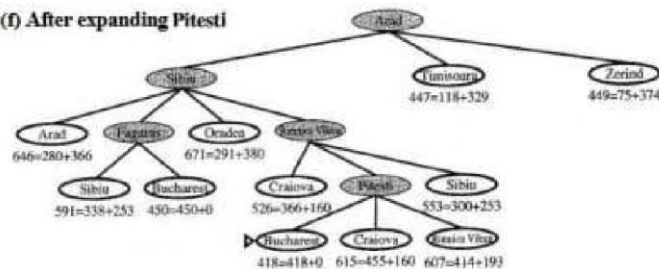
(c) After expanding Sibiu



(e) After expanding Fagaras



(f) After expanding Pitesti



UNIT IV

4/6 Marks:

1.List and explain the different components of Machine learning.?

Data storage

Facilities for storing and retrieving huge amounts of data are an important component of the learning process. Humans and computers alike utilize data storage as a foundation for advanced reasoning.

- In a human being, the data is stored in the brain and data is retrieved using electrochemical signals.
- Computers use hard disk drives, flash memory, random access memory and similar devices to store data and use cables and other technology to retrieve data.

Abstraction

The second component of the learning process is known as abstraction. Abstraction is the process of extracting

- knowledge about stored data. This involves creating general concepts about the data as a whole. The creation of knowledge involves application of known models and creation of new models.
- The process of fitting a model to a dataset is known as training. When the model has been trained, the data is transformed into an abstract form that summarizes the original information.

Generalization

The third component of the learning process is known as generalisation. The term generalization describes the process of turning the knowledge about stored data into a form that can be utilized for future action.

- These actions are to be carried out on tasks that are similar, but not identical, to those what have been seen before.
- In generalization, the goal is to discover those properties of the data that will be most relevant to future tasks.

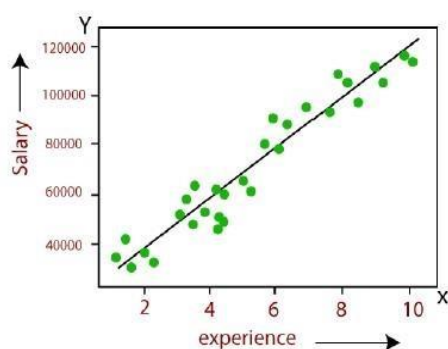
Evaluation

Evaluation is the last component of the learning process.

- It is the process of giving feedback to the user to measure the utility of the learned knowledge.
- This feedback is then utilized to effect improvements in the whole learning process.

10. Briefly explain Linear Regression. ?

- Linear regression is a statistical regression method which is used for predictive analysis.
- It is one of the very simple and easy algorithms which works on regression and shows the relationship between the continuous variables.
- It is used for solving the regression problem in machine learning.
- Linear regression shows the linear relationship between the independent variable (X-axis) and the dependent variable (Yaxis), hence called linear regression.
- If there is only one input variable (x), then such linear regression is called **simple linear regression**. And if there is more than one input variable, then such linear regression is called **multiple linear regression**.
- The relationship between variables in the linear regression model can be explained using the below image. Here we are predicting the salary of an employee on the basis of **the year of experience**.



Below is the mathematical equation for Linear regression:

$$Y = aX + b$$

Here,

Y = dependent variables (target variables), X = Independent variables (predictor variables), a and b are the linear coefficients

Some popular applications of linear regression are:

- Analyzing trends and sales estimates
- Salary forecasting
- Real estate prediction
- Arriving at ETAs in traffic.

In non-linear regression, the data can't be modelled to fit a straight line i.e. modelled to fit a curve as shown in below figure.

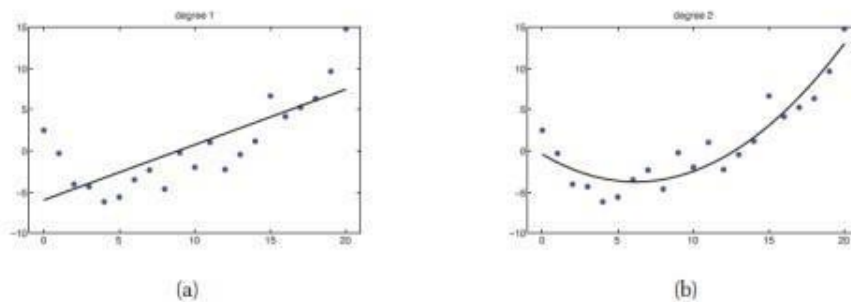


Figure (a) Linear regression on some 1d data. (b) Same data with polynomial regression (degree 2).

For example the equation can have the form: $X\theta$

Here are some examples of real-world regression problems.

- Predict tomorrow's stock market price given current market conditions and other possible side information.
- Predict the age of a viewer watching a given video on YouTube.

15. List and explain the types of Artificial Neural Network. ?

There are various types of Artificial Neural Networks (ANN) depending upon the human brain neuron and network functions, an artificial neural network similarly performs tasks. The majority of the artificial neural networks will have some similarities with a more complex biological partner and are very effective at their expected tasks. For example, segmentation or classification.

- **Feedback ANN:** In this type of ANN, the output returns into the network to accomplish the best-evolved results internally. As per the University of Massachusetts, Lowell Centre for Atmospheric Research. The feedback networks feed information back into itself and are well suited to solve optimization issues. The Internal system error corrections utilize feedback ANNs.
- **Feed-Forward ANN:** A feed-forward network is a basic neural network comprising of an input layer, an output layer, and at least one layer of a neuron. Through assessment of its output by reviewing its input, the intensity of the network can be noticed based on group behavior of the associated neurons, and the output is decided. The primary advantage of this network is that it figures out how to evaluate and recognize input patterns.

• Prerequisite

No specific expertise is needed as a prerequisite before starting this tutorial.

• Audience

Our Artificial Neural Network Tutorial is developed for beginners as well as professionals, to help them understand the basic concept of ANNs.

22. Explain the phases of NLP. ?

There are the following five phases of NLP:

1. Lexical Analysis and Morphological: The first phase of NLP is the Lexical Analysis. This phase scans the source code as a stream of characters and converts it into meaningful lexemes. It divides the whole text into paragraphs, sentences, and words.

2. Syntactic Analysis (Parsing): Syntactic Analysis is used to check grammar, word arrangements, and shows the relationship among the words.

Example: Agra goes to the Poonam

In the real world, Agra goes to the Poonam, does not make any sense, so this sentence is rejected by the Syntactic analyzer.

3. Semantic Analysis: Semantic analysis is concerned with the meaning representation. It mainly focuses on the literal meaning of words, phrases, and sentences.

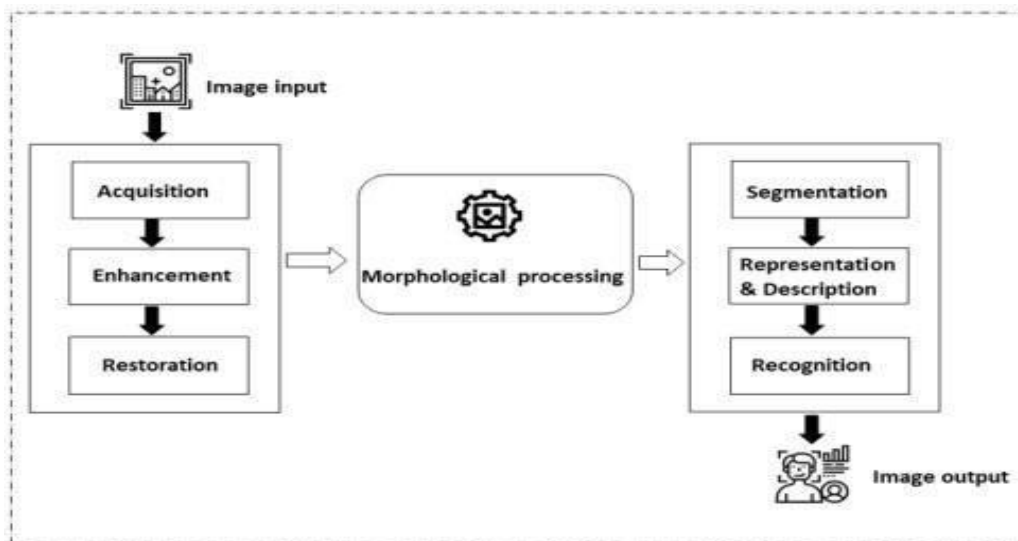
4. Discourse Integration: Discourse Integration depends upon the sentences that proceeds it and also invokes the meaning of the sentences that follow it.

5. Pragmatic Analysis: Pragmatic is the fifth and last phase of NLP. It helps you to discover the intended effect by applying a set of rules that characterize cooperative dialogues.

For Example: "Open the door" is interpreted as a request instead of an order.

28. Explain the elements of Image processing.?

AI image processing works through a combination of advanced algorithms, neural networks, and data processing to analyze, interpret, and manipulate digital images. Here's a simplified overview of how AI image processing works:



Acquisition of image: The initial level begins with image preprocessing which uses a sensor to capture the image and transform it into a usable format.

Enhancement of image: Image enhancement is the technique of bringing out and emphasising specific interesting characteristics which are hidden in an image.

Restoration of image: Image restoration is the process of enhancing an image's look. Picture restoration, as opposed to image augmentation, is carried out utilising specific mathematical or probabilistic models.

Colour image processing: A variety of digital colour modelling approaches such as HSI (Hue-Saturation-Intensity), CMY (CyanMagenta-Yellow) and RGB (Red-Green-Blue) etc. are used in colour picture processing.

Compression and decompression of image: This enables adjustments to image resolution and size, whether for image reduction or restoration depending on the situation, without lowering image quality below a desirable level. Lossy and lossless compression techniques are the two main types of image file compression which are being employed in this stage.

Morphological processing: Digital images are processed depending on their shapes using an image processing technique known as morphological operations. The operations depend on the pixel values rather than their numerical values, and well suited for the processing of binary images. It aids in removing imperfections for structure of the image.

Segmentation, representation and description: The segmentation process divides a picture into segments, and each segment is represented and described in such a way that it can be processed further by a computer. The image's quality and regional characteristics are covered by representation. The description's job is to extract quantitative data that helps distinguish one class of items from another.

Recognition of image: A label is given to an object through recognition based on its description. Some of the often-employed algorithms in the process of recognising images include the Scaleinvariant Feature Transform (SIFT), the Speeded Up Robust Features (SURF), and the PCA (Principal Component Analysis)

33.Explain the different types of Robots.?

1. Androids

Androids are robots that resemble humans. They are often mobile, moving around on wheels or a track drive. According to the American Society of Mechanical Engineers, these humanoid robots are used in areas such as caregiving and personal assistance, search and rescue, space exploration and research, entertainment and education, public relations and healthcare, and manufacturing.

2. Telechir

A telechir is a complex robot that is remotely controlled by a human operator for a telepresence system. It gives that individual the sense of being on location in a remote, dangerous or alien environment, and enables them to interact with it since the telechir continuously provides sensory feedback.

3. Telepresence robot

A telepresence robot simulates the experience -- and some capabilities -- of being physically present at a location. It combines remote monitoring and control via telemetry sent over radio, wires or optical fibers, and enables remote business consultations, healthcare, home monitoring, childcare and more.

4. Industrial robot

The IFR (International Federation of Robotics) defines an industrial robot as an "automatically controlled, reprogrammable multipurpose manipulator programmable in three or more axes." Users can adapt these robots to different applications as well. Combining these robots with AI has helped businesses move them beyond simple automation to higher-level and more complex tasks.

5.Swarm robot

Swarm robots (aka insect robots) work in fleets ranging from a few to thousands, all under the supervision of a single controller. These robots are analogous to insect colonies, in that they exhibit

simple behaviors individually, but demonstrate behaviors that are more sophisticated with an ability to carry out complex tasks in total.

6. Smart robot

This is the most advanced kind of robot. The smart robot has a built-in AI system that learns from its environment and experiences to build knowledge and enhance capabilities to continuously improve. A smart robot can collaborate with humans and help solve problems in areas like the following:

- agricultural labor shortages;
- food waste;
- study of marine ecosystems;

UNIT III

4/6Marks:

6. Consider given KB and using inference rule make the conclusion neither [1,2] nor [2,1] contains pit.?

The KB consists of the following statements:

1. $R1 : \neg P_{1,1}$ (There is no pit in [1, 1])
2. $R2 : B_{1,1} \leftrightarrow (P_{1,2} \vee P_{2,1})$ (The breeze in [1, 1] is equivalent to the existence of a pit in [1, 2] or [2, 1])
3. $R3 : B_{2,1} \leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$ (The breeze in [2, 1] is equivalent to the existence of a pit in [1, 1], [2, 2], or [3, 1])
4. $R4 : \neg B_{1,1}$ (There is no breeze in [1, 1])
5. $R5 : B_{2,1}$ (There is a breeze in [2, 1])

We need to conclude that neither [1, 2] nor [2, 1] contains a pit.

Step-by-Step Inference:

1. From $R4: \neg B_{1,1}$
Since there is no breeze in [1, 1].

2. Use $R2$ with $\neg B_{1,1}$:
 $B_{1,1} \leftrightarrow (P_{1,2} \vee P_{2,1})$

Since $\neg B_{1,1}$ is true, this implies:

$$\neg(P_{1,2} \vee P_{2,1})$$

Which simplifies to:

$$\neg P_{1,2} \wedge \neg P_{2,1}$$

Thus, we can conclude:

$$\neg P_{1,2} \text{ and } \neg P_{2,1}$$

This means that there are no pits in [1, 2] and [2, 1].

8. Explain the various logical connectives used in propositional logic.?

1. $\neg$ (Not) Negation:

- **Symbol:** $\neg$
- **Meaning:** Negation inverts the truth value of a given proposition. If a proposition P is true, then its negation $\neg P$ will be false, and if P is false, then $\neg P$ will be true. This connective is crucial for expressing the opposite of a given statement.

2. $\wedge$ (And) Conjunction:

- **Symbol:** $\wedge$
- **Meaning:** Conjunction is used to combine two propositions and is true only if both propositions are true. For $P \wedge Q$ (P and Q), the statement is true if both P and Q are true; otherwise, it is false. This connective is fundamental for expressing conditions that require multiple criteria to be met simultaneously.

3. $\vee$ (Or) Disjunction:

- **Symbol:** $\vee$
- **Meaning:** Disjunction represents an inclusive "or" and is true if at least one of the propositions is true. For $P \vee Q$ (P or Q), the statement is true if P is true, Q is true, or both are true. It is only false when both P and Q are false. This connective is useful for scenarios where multiple alternatives are acceptable.

4. $\rightarrow$ (Implies):

- **Symbol:** $\rightarrow$
- **Meaning:** Implication describes a conditional relationship between two propositions. For $P \rightarrow Q$ (If P then Q), the statement is true in all cases except when P is true and Q is false. This means that if P happens, then Q must also happen; otherwise, the implication holds true. This connective is essential for expressing dependencies and causations.

5. $\leftrightarrow$ (Double implication):

- **Symbol:** $\leftrightarrow$
- **Meaning:** Double implication, or bi-conditional, indicates that both propositions must have the same truth value for the statement to be true. For $P \leftrightarrow Q$ (P if and only if Q), the statement is true if both P and Q are either true or false. This connective is used to express equivalence between propositions, where one is true if and only if the other is true.

2. Explain entailment in wumpus world problem with example?

Entailment

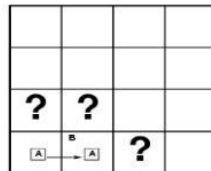
- Entailment means that one thing **follows from** another:

$$KB \models \alpha$$

- Knowledge base KB entails sentence α if and only if α is true in all worlds where KB is true
 - If α true then KB must also be true
 - E.g., the KB containing “the Giants won” and “the Reds won” entails “Either the Giants won or the Reds won”
 - E.g., $x+y = 4$ entails $4 = x+y$
 - Entailment is a relationship between sentences (i.e., **syntax**) that is based on **semantics**

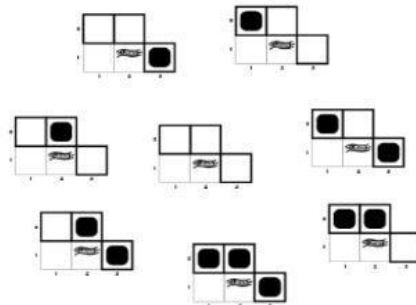
Entailment in the wumpus world

- Situation after detecting nothing in [1,1], moving right, breeze in [2,1]

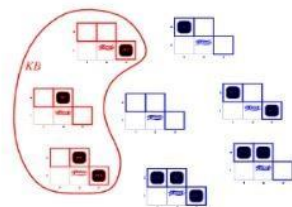


3 Boolean choices $\Rightarrow$ 8 possible models

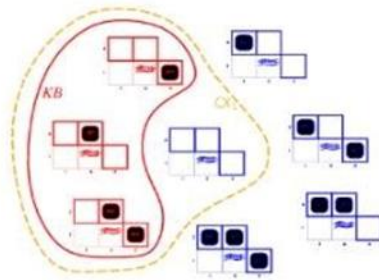
for the adjacent squares [1,2], [2,2] and [3,1] to contain pits



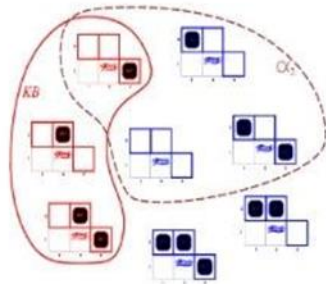
Consider possible models for KB assuming only pits



- KB = wumpus-world rules + observations
- KB is false in any model in which [1,2] contains a pit, because there is no breeze in [1,1]



- Consider $\alpha_1 = "[1,2] \text{ is safe}" = "There is no pit in [1,2]"$
- In every model KB is true α_1 is also true
- $KB \models \alpha_1$, proved by model checking
- We can conclude that there is no pit in [1,2]



- Consider $\alpha_2 = "[2,2] \text{ is safe}" = "There is no pit in [2,2]"$
- In some models in which KB is true α_2 is false
- $KB \not\models \alpha_2$
- We cannot conclude that there is no pit in [2,2]

18.Explain Forward-chaining algorithm with example in propositional logic.?

The best way to understand the algorithm is through an example and a picture. Figure 7.15(a) shows a simple knowledge base of Horn clauses with A and B as known facts. ANWR GRAPH Figure 7.15(b) shows the same knowledge base drawn as an AND-OR graph. In

AND-OR graphs, multiple links joined by an arc indicate a conjunction--every link must be proved while multiple links without an arc indicate a disjunction-any link can be proved. It is easy to see how forward chaining works in the graph. The known leaves (here, A and B) are set, and inference propagates up the graph as far as possible. Wherever a conjunction appears, the propagation waits until all the conjuncts are known before proceeding. The reader is encouraged to work through the example in detail.

```

function PL-FC-ENTAILS?(KB, q) returns true or false
inputs: KB, the knowledge base, a set of propositional Horn clauses
         q, the query, a proposition symbol
local variables: count, a table, indexed by clause, initially the number of premises
                   inferred, a table, indexed by symbol, each entry initially false
                   agenda, a list of symbols, initially the symbols known to be true in KB

while agenda is not empty do
  p ← POP(agenda)
  if p = q then return true
  unless inferred[p] do
    inferred[p] ← true
    for each Horn clause c in whose premise p appears do
      decrement count[c]
      if count[c] = 0 then
        PUSH(HEAD[c], agenda)
return false

```

Figure 7.14 The forward-chaining algorithm for propositional logic. The *agenda* keeps track of symbols known to be true but not yet "processed." The *count* table keeps track of how many premises of each implication are as yet unknown. Whenever a new symbol *p* from the agenda is processed, the count is reduced by one for each implication in whose premise *p* appears. (These can be identified in constant time if *KB* is indexed appropriately.) If a count reaches zero, all the premises of the implication are known so its conclusion can be added to the agenda. Finally, we need to keep track of which symbols have been processed; an inferred symbol need not be added to the agenda if it has been processed previously. This avoids redundant work; it also prevents infinite loops that could be caused by implications such as $P \Rightarrow Q$ and $Q \Rightarrow P$.

It is easy to see that forward chaining is sound: every inference is essentially an application of Modus Ponens. Forward chaining is also complete: every entailed atomic sentence will be derived. The easiest way to see this is to consider the final state of the inferred table (after the algorithm reaches a fixed point where no new inferences are possible). The table contains true for each symbol inferred during the process, and false for all other symbols. We can view the table as a logical model; moreover, every definite clause in the original KB is true in this model. To see this, assume the opposite, namely that some clause $a_1 A_1 \dots A_k \rightarrow b$ is false in the model. Then $a_1 A_1 \dots A_k$ must be true in the model and b must be false in the model. But this contradicts our assumption that the algorithm has reached a fixed point! We can conclude, therefore, that the set of atomic sentences inferred at the fixed point defines a model of the original KB. Furthermore, any atomic sentence *q* that is entailed by the KB must be true in all its models and in this model in particular. Hence, every entailed sentence *q* must be inferred by the algorithm.

Forward chaining is an example of the general concept of data-driven reasoning—that is, reasoning in which the focus of attention starts with the known data. It can be used within an agent to derive conclusions from incoming percepts, often without a specific query in mind. For example, the wumpus agent might TELL its percepts to the knowledge base using an incremental forward-chaining algorithm in which new facts can be added to the agenda to initiate new inferences. In humans, a certain amount of data-driven reasoning occurs as new information arrives. For example, if I am indoors and hear rain starting to fall, it might occur to me that the picnic will be canceled. Yet it will probably not occur to me that the seventeenth petal on the largest rose in my neighbor's garden will get wet; humans keep forward chaining under careful control, lest they be swamped with irrelevant consequences.

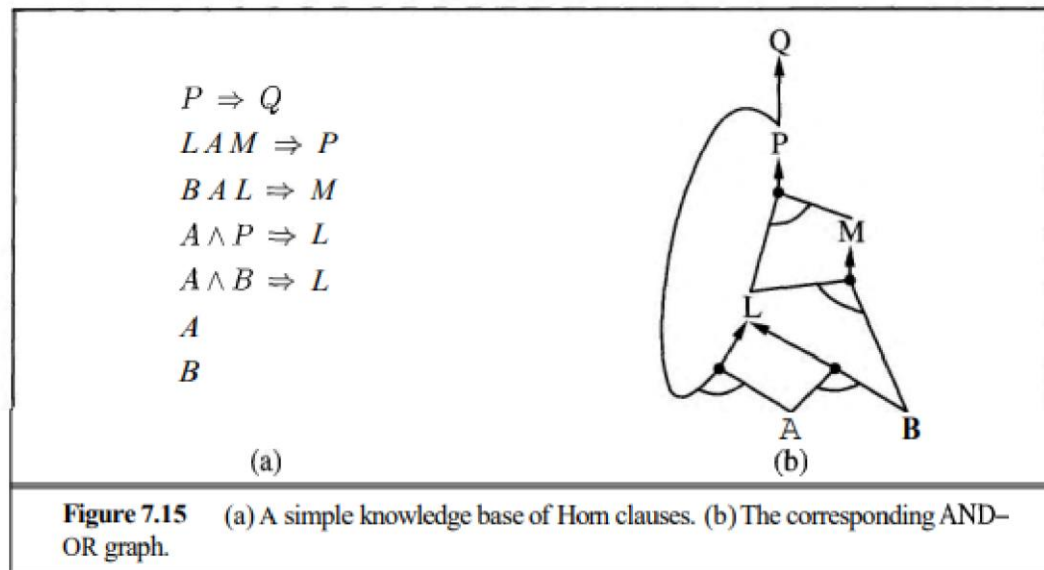


Figure 7.15 (a) A simple knowledge base of Horn clauses. (b) The corresponding AND-OR graph.

24. Explain model for first order logic with example.?

- ❖ First-order logic is also known as Predicate logic or First-order predicate logic.
- ❖ First-order logic, like natural language has well defined syntax and semantics.
- ❖ It assumes the world contains
 - Objects: people, houses, numbers, colors, baseball games, wars
 - Relations: red, round, prime, brother of, bigger than, part of, comes between,
 - Functions: father of, best friend, one more than, plus,....

Properties of FOL

- ❖ It has ability to represent facts about some or all of the objects and relations in the universe
- ❖ Represent law and rules extracted from real world
- ❖ Useful language for Maths, Philosophy and AI
- ❖ Represent facts in realistic manner rather than just true / false * Makes ontological commitment

First-order logic

Where as propositional logic assumes the world contains facts, first-order logic (like natural language) assumes the world contains,

- Objects: people, houses, numbers, colors, baseball games, wars,
- Relations: red, round, prime, brother of, bigger than, part of, comes between
- Functions: father of, best friend, one more than, plus, (relations in which there is only one value for a given input)

33. Explain backward chaining algorithm in FOL.?

Backward chaining in First-Order Logic (FOL), also known as backward reasoning, is a method used for automated theorem proving and reasoning in artificial intelligence and logic programming.

Figure 9.6 shows a simple backward-chaining algorithm, FOL-BC-ASK. It is called with a list of goals containing a single element, the original query, and returns the set of all substitutions satisfying the query. The list of goals can be thought of as a "stack" waiting to be

worked on; if all of them can be satisfied, then the current branch of the proof succeeds. The algorithm takes the first goal in the list and finds every clause in the knowledge base whose positive literal, or

head, unifies with the goal. Each such clause creates a new recursive call in which the premise, or body, of the clause is added to the goal stack. Remember that facts are clauses with a head but no body, so when a goal unifies with a known fact, no new subgoals

are added to the stack and the goal is solved. Figure 9.7 is the proof tree for deriving *Criminal*(West) from sentences (9.3) through (9.10).

```

function FOL-BC-ASK(KB, goals,  $\theta$ ) returns a set of substitutions
  inputs: KB, a knowledge base
           goals, a list of conjuncts forming a query ( $\theta$  already applied)
            $\theta$ , the current substitution, initially the empty substitution {}
  local variables: answers, a set of substitutions, initially empty

  if goals is empty then return { $\theta$ }
   $q' \leftarrow \text{SUBST}(\theta, \text{FIRST}(\text{goals}))$ 
  for each sentence r in KB where  $\text{STANDARDIZE-APART}(r) = (p_1 \wedge \dots \wedge p_n \Rightarrow q)$ 
    and  $\theta' \leftarrow \text{UNIFY}(q, q')$  succeeds
       $\text{new-goals} \leftarrow [p_1, \dots, p_n | \text{REST}(\text{goals})]$ 
       $\text{answers} \leftarrow \text{FOL-BC-ASK}(\text{KB}, \text{new-goals}, \text{COMPOSE}(\theta', \theta)) \cup \text{answers}$ 
  return answers

```

Figure 9.6 A simple backward-chaining algorithm.

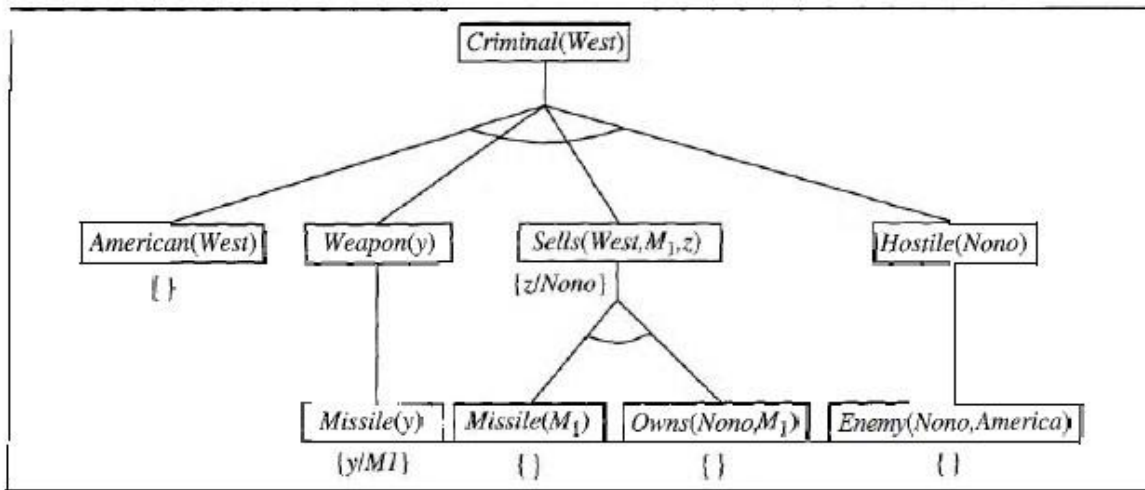


Figure 9.7 Proof tree constructed by backward chaining to prove that West is a criminal. The tree should be read depth first, left to right. To prove *Criminal*(West), we have to prove the four conjuncts below it. Some of these are in the knowledge base, and others require further backward chaining. Bindings for each successful unification are shown next to the corresponding subgoal. Note that once one subgoal in a conjunction succeeds, its substitution is applied to subsequent subgoals. Thus, by the time *FOL-BC-ASK* gets to the last conjunct, originally *Hostile*(*z*), *z* is already bound to *Nono*.

The algorithm uses **composition** of substitutions. $\text{COMPOSE}(\theta_1, \theta_2)$ is the substitution whose effect is identical to the effect of applying each substitution in turn. That is,

$$\text{SUBST}(\text{COMPOSE}(\theta_1, \theta_2), p) = \text{SUBST}(\theta_2, \text{SUBST}(\theta_1, p)) .$$

In the algorithm, the current variable bindings, which are stored in θ , are composed with the bindings resulting from unifying the goal with the clause head, giving a new set of current bindings for the recursive call.

Example of backward chaining

- The information provided in the previous example (forward chaining) can be used to provide a simple explanation of backward chaining. Backward chaining can be explained in the following sequence.
- B
- $A \rightarrow B$
- A
- B is the goal or endpoint, that is used as the starting point for backward tracking. A is the initial state. $A \rightarrow B$ is a fact that must be asserted to arrive at the endpoint B.
- A practical example of backward chaining will go as follows:
- Tom is sweating (B).
- If a person is running, he will sweat ($A \rightarrow B$).
- Tom is running (A).

9

32. Explain forward chaining algorithm in FOL. ?

Forward chaining is a data-driven reasoning method used in artificial intelligence, particularly in rule-based systems, for inferring new facts from known facts and rules. In first-order logic (FOL), forward chaining uses a set of inference rules to derive conclusions from given premises.

Forward Chaining Algorithm Steps

1. Initialization:

- Start with a set of known facts (initial database).
- Initialise the working memory with these facts.

2. Rule Matching:

- Identify rules whose premises match the facts in the working memory.
- In FOL, this involves unifying the premises with the facts.

3. Rule Application:

- Select and apply a matching rule.
- Infer new facts from the rule's conclusion and add them to the working memory.

4. Iteration:

- Repeat rule matching and application until:
 - A goal fact is derived, or
 - No more rules can be applied.

5. Termination:

- The process stops when the goal fact is derived or no new inferences can be made.

Example Facts:

- i. American(West)American(West)American(West)
- ii. Missile(M1)Missile(M_1)Missile(M1)
- iii. Owns(Nono,M1)Owns(Nono, M_1)Owns(Nono,M1)
- iv. Sells(West,M1,Nono)Sells(West, M_1, Nono)Sells(West,M1 ,Nono)
- v. Enemy(Nono,America)Enemy(Nono, America)Enemy(Nono,America)

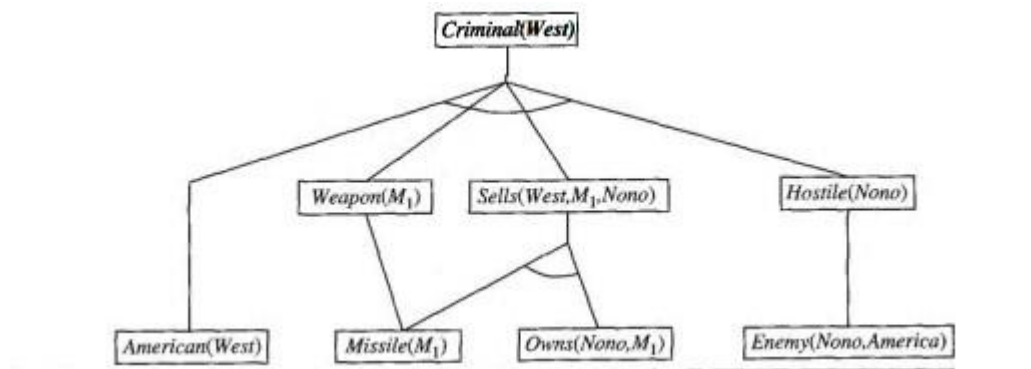
Example Rules:

- i. $\text{American}(x) \wedge \text{Weapon}(y) \wedge \text{Sells}(x,y,z) \wedge \text{Hostile}(z) \rightarrow \text{Criminal}(x)$
- ii. $\text{Missile}(x) \rightarrow \text{Weapon}(x)$
- iii. $\text{Enemy}(x,\text{America}) \rightarrow \text{Hostile}(x)$

Query:

- Is $\text{Criminal}(\text{West})$ true?

Diagram:



Pseudocode:

```
def forward_chaining(facts, rules, goal):
```

```
    working_memory = set(facts)
```

```
    while True:
```

```
        new_facts = set()
```

```
    for rule in rules:
```

```
        if rule.premise_matches(working_memory):
```

```
            inferred_fact = rule.apply()
```

```
        if inferred_fact == goal:
```

```
            return True
```

```
    new_facts.add(inferred_fact)
```

```
    if not new_facts:
```

```
    return False
```

```
    working_memory.update(new_facts)
```

```
Algorithm Steps Applied to Example:
```

1. Initialization:

- Facts in KB:

{American(West),Missile(M1),Owns(Nono,M1),Sells(West,M1,Nono),Enemy (Nono,America)}

2. First Iteration:

- Apply Rule 2: Missile(M1)→Weapon(M1)
- New fact inferred: Weapon(M1)

3. Second Iteration:

- Apply Rule 3: Enemy(Nono,America)→Hostile(Nono)
- New fact inferred: Hostile(Nono)

4. Third Iteration:

- Apply Rule 1:

American(West)∧Weapon(M1)∧Sells(West,M1,Nono)∧Hostile(Nono)→Criminal(West)

- All premises match known facts.
- New fact inferred: Criminal(West)

Result:

- The query Criminal(West) is true as it has been inferred through forward chaining.

Summary

Forward chaining in FOL involves initialising with known facts, matching rules with these facts, applying rules to infer new facts, and iterating this process until the query is confirmed or no new facts are generated. In the given example, forward chaining successfully infers Criminal(West) based on the initial facts and rules in the knowledge base.

2.Explain PEAS of wumpus world problem.?

The PEAS (Performance measure, Environment, Actuators, Sensors) framework is a conceptual model used in artificial intelligence to describe the key components of an intelligent agent and its interaction with the environment. When applied to the Wumpus World problem, the PEAS framework can be described as follows:

Performance Measure:

-The performance measure in the Wumpus World problem is the evaluation of the agent's behavior in the environment. It quantifies how well the agent is doing in achieving its goals. In the case of the Wumpus World, the performance measure could be based on factors such as the number of safe rooms explored, the number of gold pieces collected, and the number of arrows used to kill the Wumpus. The goal is to maximize the agent's score by safely navigating the environment and collecting the gold while avoiding hazards.

Environment:

-The environment in the Wumpus World problem is a grid-based cave system with pits, a Wumpus (a dangerous creature), gold, and the agent itself. The environment is dynamic, as the agent's actions can change the state of the environment. The agent must navigate through the environment, avoid hazards, and collect the gold while dealing with uncertainty about the location of the Wumpus and pits.

Actuators:

-The actuators are the actions that the agent can perform to affect the environment. In the Wumpus World, the agent can move forward, turn left or right, grab the gold, shoot an arrow to kill the Wumpus, and climb out of the cave. These actions allow the agent to explore the environment, collect rewards, and mitigate risks.

Sensors:

-The sensors provide the agent with information about the state of the environment. In the Wumpus World, the agent has sensors to perceive whether there is a breeze (indicating a pit nearby), a stench (indicating the presence of the Wumpus), a glitter (indicating gold), and a bump (indicating a wall). These sensory inputs help the agent make decisions about its actions and navigate the environment safely.

Example:

-Suppose the agent's performance measure is based on the number of gold pieces collected, minus penalties for falling into pits or being killed by the Wumpus. The environment consists of a 4x4 grid with pits, a Wumpus, gold, and the agent. The actuators allow the agent to move, grab gold, shoot arrows, and climb out of the cave. The sensors provide information about the presence of hazards and rewards in adjacent squares.